Wi-Fi Hacking (Theory part)

In this topic, we learn about wireless networking concepts, standards, and encryption protocols like WEP, WPA, and WPA3. It also covers common wireless vulnerabilities and attack methods such as eavesdropping, spoofing, and deauthentication used in penetration testing.

Wireless concepts

- There are a number of different wireless technologies out there, but when it comes to wireless LANS (WLANs), understand that your WLAN uses radio frequencies that travel through the air as radio waves.

- This means you don't need a direct line of sight to the system you're communicating with, since radio waves can travel through walls unlike infrared waves, which require a clear line of sight.

Wireless Agencies

- Wireless communication and its surrounding technologies are governed by a number of different agencies.

- List of some of the popular wireless agencies to be familiar with:
    - IEEE: [Institute of Electrical and Electronics Engineers] The IEEE is responsible for creating the different wireless networking standards, such as 802.11b, 802.11g, 802.11n, and 802.11ac. You can learn more about the IEEE at www.ieee.org.

    - Wi-Fi Alliance: The Wi-Fi Alliance is responsible for ensuring compatibility and interoperability of wireless networking components. The Wi-Fi Alliance is responsible for testing and certifying components to be Wi-Fi compatible. You can learn more about the Wi-Fi Alliance at www.wi-fi.org.

    - Federal Communications Commission: The Federal Communications Commission (FCC) is a U.S. federal agency that has the responsibility of regulating the use of wireless devices and frequencies. The FCC defined three frequency ranges that are for public use: 900 MHz, 2.4 GHz, and 5 GHz.

Rushi 20Pute

Wireless LAN frequencies

- Wireless components are designed to use a frequency range. The most popular frequency range for wireless networks is the 2.4 GHz range.

- This frequency range is divided into a number of different channels, with each channel running on a different 22 MHz frequency within the 2.4 GHz frequency range.

- Three frequency ranges are available for public use as defined by the FCC: the 900 MHz range, the 2.4 GHz range, and the 5 GHz range.

2.4 GHz Frequency Ranges

Channel	Frequency Range
1	2.3995 GHz – 2.4245 GHz
2	2.4045 GHz – 2.4295 GHz
3	2.4095 GHz – 2.4345 GHz
4	2.4145 GHz – 2.4395 GHz
5	2.4195 GHz – 2.4445 GHz
6	2.4245 GHz – 2.4495 GHz
7	2.4295 GHz – 2.4545 GHz
8	2.4345 GHz – 2.4595 GHz
9	2.4395 GHz – 2.4645 GHz
10	2.4445 GHz – 2.4695 GHz
11	2.4495 GHz – 2.4745 GHz
12	2.4545 GHz – 2.4795 GHz
13	2.4595 GHz – 2.4845 GHz

Rushi2Pute

Wireless equipment and configuration

- **Wireless network card**
  To connect to the wireless network, clients will need a wireless network card installed in their systems. As a penetration tester, you will need to ensure that you have a wireless network card supported by Kali Linux. The wireless card should also support packet injection. Packet injection is needed to be able to monitor all traffic on the wireless network and inject packets onto the network. You must place the wireless adapter in monitor mode in order to do this.

- **Wireless access point**
  The next wireless network component to mention is a popular device known as a wireless access point. The wireless access point is a device that has antennas to send and receive the wireless traffic as well as a network port to connect the access point to the wired network. This allows systems on the wireless network to access the wired network and vice versa.

- **The SSID**
  The wireless access point is configured with a service set identifier (SSID), which acts as a name for the wireless network. In order for wireless clients to connect to the wireless access point, they must specify the name of the wireless network, or SSID. This is not much of a security feature as the SSID is broadcasted onto the network by default.

- **Wireless clients**
  Wireless clients are systems that have a wireless network card installed and have been configured to connect to the wireless access point or even to other wireless clients directly. When a wireless client accesses a device on the wired network, such as a printer or server, the information is sent to the wireless access point from the wireless client and then onto the wired network.

Rushi 20Pute

Wireless Standards

- The IEEE is responsible for defining networking standards including the standards that govern wireless networking.

- Wireless networking falls under the IEEE 802.11 project model. A number of different wireless standards have been developed over the years is as follows:

    - **802.11a**
      The first major wireless networking standard is known as the 802.11a network standard. 802.11a runs at 54 Mbps and uses the 5 GHz frequency range as opposed to the popular 2.4 GHz range. Wireless devices that are 802.11a devices are incompatible with 802.11b and 802.11g devices.

    - **802.11b**
      The 802.11b wireless standard runs at 11 Mbps and uses the 2.4 GHz frequency range. Because 802.11b uses a different frequency range than the 802.11a wireless standard, they are incompatible with one another. For example, you could not have a client using an 802.11a network card connect to an 802.11b access point. The 802.11b standard became the Wi-Fi standard, and as future standards and technologies are developed, they will be part of the Wi-Fi standard as well.

    - **802.11g**
      The 802.11g wireless standard follows the Wi-Fi standard and is therefore Compatible with the 802.11b wireless standard. 802.11g runs at 54 Mbps and uses the 2.4 GHz frequency that is used by 802.11b (which is one of the reasons why they are compatible)

    - **802.11n**
      802.11n came out around 2009 and is compatible with the 802.11a, 802.11b, and 802.11g wireless standards because it can run at either the 2.4 GHz frequency or the 5 GHz frequency. 802.11n has a number of new features that help increase transfer rate, including the use of multiple antennas and a feature called channel bonding, which allows the device to send data over multiple channels.
      The 802.11n transfer rate depends on the mode of the channel (20 MHz or 40 MHz) and the number of antennas used. Using 40 MHz channels will give you a higher transfer rate, and using more antennas will increase the transfer rate. For example, 802.11n using four antennas in 20 MHz mode will have a transfer rate of 150 Mbps, while using four antennas in the 40 MHz mode will have a transfer rate of 600 Mbps.

Rushi 20Pute

- **802.11ac**
  The 802.11ac wireless standard came out around 2014 and runs at the 5 GHz frequency, which is great because it means that it will not have interference with microwaves or cordless phones. The 802.11ac builds on the many performance features of the 802.11n standard and can reach transfer rates of 1 Gbps or more depending on the configuration (number of antennas and channel mode).

Wireless Network Standards

	802.11a	802.11b	802.11g	802.11n	802.11ac
Frequency	5 GHz	2.4 GHz	2.4 GHz	2.4/5 GHz	5 GHz
Transfer Rate	54 Mbps	11 Mbps	54 Mbps	>150 Mbps	>1 Gbps
Range	150 Feet	300 Feet	300 Feet	300 Feet	300 Feet
Compatibility	802.11n	802.11g/n	802.11b/n	802.11a/b/g	802.11n

Encryption protocols

Most wireless networks that have been secured are using some form of encryption protocol to encrypt the traffic between the wireless client and the wireless access point.

- **WEP**
  - The Wired Equivalent Privacy (WEP) wireless security protocol is popular with ●802.11b devices.
  - WEP uses a pre-shared key (meaning you must tell the client the key in order to connect) assigned on the access point and the client to encrypt and decrypt the traffic.
  - The pre-shared key is required at the client in order for the client to connect to the network.
  - WEP supports 64-bit or 128-bit encryption and uses the RC4 symmetric encryption protocol.
  - WEP encryption is easily cracked and should not be used. You might notice that many new wireless devices do not support WEP as an option within the configuration settings.

Rushi 20Pute

- **WPA**
  - The WEP encryption protocol has some flaws in how it performs the encryption, so the Wi-Fi Alliance created the Wi-Fi Protected Access (WPA) protocol. WPA uses the Temporal Key Integrity Protocol (TKIP), which is a protocol that changes the encryption key with every packet sent to try to make it harder for hackers to crack the key.
  - WPA consists of two modes, with each mode providing a different level of security:

    - **WPA2**
      The Wi-Fi Alliance updated the WPA protocol to WPA2 and added some improvements to the protocol, such as the fact that WPA2 uses the Advanced Encryption Standard (AES) protocol, which has yet to be cracked. AES supports 128-bit, 192-bit, and 256-bit encryption! Like WPA, WPA2 also supports both personal and enterprise modes.

    - **WPA3**
      In 2018, a new version of the WPA security protocol known as WPA3 was released. The WPA3 protocol also supports both personal mode and enterprise mode. With personal mode, WPA3 uses AES-128 (128-bit encryption) and with enterprise mode, it uses 192-bit encryption.
      The WPA3 security protocol has some major security improvements that help prevent some of the attacks that have occurred against the WPA2 protocol, including:
      - Simultaneous Authentication of Equals (SAE): One of the attacks against WPA2 involves using the handshake traffic to crack the encryption, so WPA implemented the Simultaneous Authentication of Equals (SAE) authentication method. SAE allows mutual authentication of the access point and wireless client, meaning the access point authenticates the client and the client authenticates the access point.
      - Session keys: WPA3 uses unique session keys in the encryption process so that a wireless client cannot decrypt the traffic of another.

Rushi20Pute

Exploiting Wireless Vulnerabilities

An attack method represents a general technique used to perform an attack.

- **Eavesdropping**: Many of the attacks on wireless networks involve eavesdropping on traffic flowing through the air. Eavesdropping refers to the process of capturing the traffic, making a copy of it so that it can be read if it is not encrypted, or cracking the encryption key if it is encrypted.

- **Data modification:** With wireless networks, an attacker can perform a man-in-the-middle (MiTM) attack (now known as an on-path attack), intercept the traffic, and then modify the traffic before forwarding it on to the wireless access point.

- **Data corruption:** Data corruption attack methods involve altering or corrupting the data so that it is not readable by the destination system.

- **Relay attacks:** A relay attack is when an attacker captures a wireless signal and sends it somewhere else (without really looking at it). This type of attack is commonly used to compromise vehicle key fobs. Using a special transmitter, a hacker or pen tester could relay the signal from the key fob of your car to a fellow attacker who receives the relayed signal to open the door of your car (without needing the key).

- **Spoofing:** Spoofing refers to altering a source address, whether that is the source MAC address or source IP address of a packet, or the source email address of an email message. It is common in wireless attacks that the attacker modifies its source MAC address in order to bypass MAC filtering on the wireless access point.

- **Deauthentication:** Deauthentication attacks are common because in order to crack the wireless encryption, the attacker needs to capture (through eavesdropping) the authentication traffic. If users have already authenticated to the access point, the attacker or pen tester would not be able to capture the authentication traffic. So, the attacker or pen tester deauthenticates the wireless clients, which forces them to authenticate again and enables the attacker to capture the traffic.

- **Jamming:** In this attack method, the attacker purposely blocks or interferes with the wireless transmission. This could be done to act as a denial of service (DoS) attack against the wireless network.

Rushi 20Pute

- **Capturing handshakes:** The handshake traffic that occurs during the initial connection to the wireless network can be vulnerable to cryptography attacks, so capturing the initial handshake traffic is critical to many wireless attacks.

- **On-path:** As mentioned earlier, on-path attacks were formerly known as man-in-the-middle (MiTM) attacks. So on-path attacks are critical to capturing wireless traffic so that the wireless encryption can be cracked.

- **Wireless network card:** If you are going to perform a pen test on a wireless network, you need to have a wireless network card that is detected by your operating system (typically Kali Linux) and can be placed in monitor mode. Monitor mode allows you to view all wireless network traffic.

- **Amplified antenna:** Another hardware item you may look to have in your toolset is an amplified antenna, which is a special antenna that allows you to extend the range of the wireless signal.

- **Aircrack-ng:** Aircrack-ng (www.kali.org/tools/aircrack-ng) is a suite of tools available on Kali Linux that allows you to exploit wireless networks.

  Following is a list of the tools that come with the Aircrack-ng suite:
  - Aircrack-ng: Used to crack encryption keys for WEP, WPA, and WPA2.
  - Airmon-ng: Used to place the wireless network card in monitor mode.
  - Aireplay-ng: Used to perform packet injection.
  - Airodump-ng: Used to capture wireless traffic.
  - Airbase-ng: Used to create a fake access point for a man-in-the-middle attack.

Rushi 20Pute